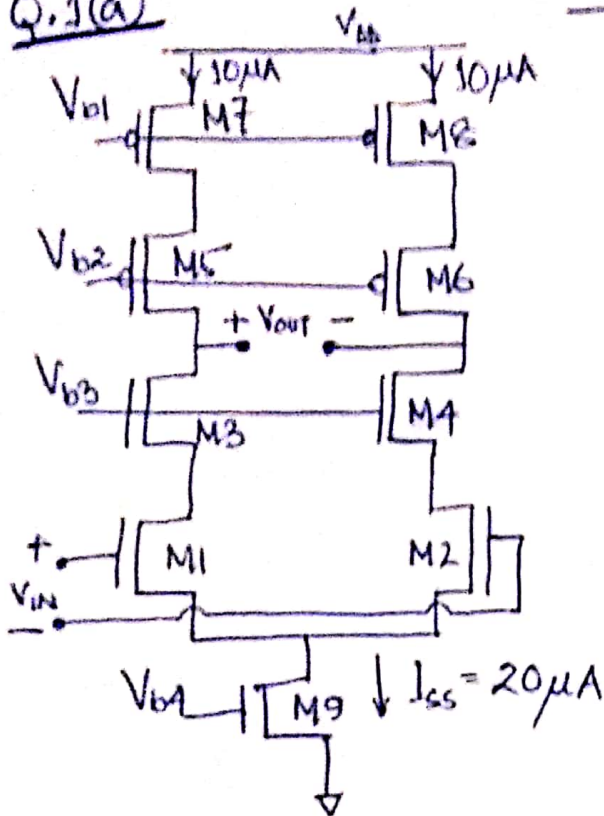


19-20 ADVD
Part A Solutions

Q.1(a)



ADVD

$$\begin{aligned} ICMR_{MAX} &= V_{DD} - V_{SD7} - V_{SD5} - V_{DS3} + V_{TH1} \\ &= 2.5 - \sqrt{\frac{2 \times 10}{40 \times 10}} - \sqrt{\frac{2 \times 10}{40 \times 10}} - \sqrt{\frac{2 \times 10}{120 \times 10}} + 0.4 \end{aligned}$$

$$= \boxed{2.323 \text{ V}} \quad (2)$$

$$ICMR_{MIN} = V_{TH1} + V_{OV1} + V_{OV9}$$

$$= 0.4 + \sqrt{\frac{2 \times 10}{120 \times 10}} + \sqrt{\frac{2 \times 20}{120 \times 10}}$$

$$= \boxed{0.711 \text{ V}} \quad (2)$$

Small Signal differential Gain :-

$$\text{Diff. Gain} \approx g_{m1} [g_{m4} r_{o4} r_{o2} \parallel g_{m6} r_{o6} r_{o8}]$$

$$g_{m1} = \frac{2I_{D,M1}}{V_{OV1}} = \frac{2 \times 10}{0.129} \mu\text{A/V} = 155 \mu\text{A/V}$$

$$g_{m4} = g_{m1} = 155 \mu\text{A/V}$$

$$g_{m6} = \frac{2 \times I_{D,M6}}{V_{OV6}} = \frac{20}{0.223} \mu\text{A/V} = 89.67 \mu\text{A/V}$$

$$r_{o2} = r_{o4} = \frac{1}{\lambda_N I_D} = \frac{1}{0.05 \times 10 \times 10^{-6}} \Omega = 2 \text{ M}\Omega$$

$$r_{o6} = r_{o8} = \frac{1}{\lambda_P I_D} = \frac{1}{0.05 \times 10 \times 10^{-6}} \Omega = 2 \text{ M}\Omega$$

$$V_{OV} = \sqrt{\frac{2I_D}{\mu C_{ox} \left(\frac{W}{L}\right)}}$$

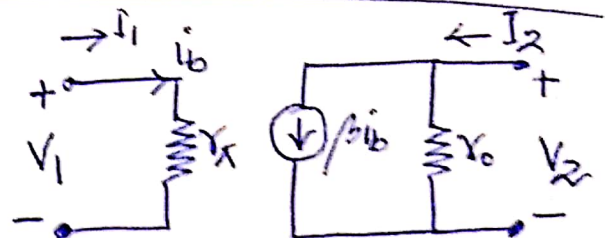
$$\therefore g_{m4} r_{o4} r_{o2} \parallel g_{m6} r_{o6} r_{o8} = (620 \times 10^6) \parallel (358.68 \times 10^6) \\ = 227.22 \times 10^6 \quad (2)$$

$$\therefore \text{Diff. Gain} = 155 \times 10^{-6} \times 227.22 \times 10^6 = \boxed{35220.04} \quad (2)$$

Output Voltage Swing :-

$$\begin{aligned} \text{O/P Swing} &= 2 \left[(V_{DD} - V_{ov8} - V_{ov6}) - (V_{ov4} + V_{ov2} + V_{ov9}) \right] \\ &= 2 \left[\{2.5 - (2 \times 0.223)\} - (0.129 + 0.129 + 0.182) \right] \text{ V} \\ &= 2 \times 1.614 \text{ V} = \boxed{3.228 \text{ V}} \quad (2) \end{aligned}$$

Q.1(b)
$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$



$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \text{--- (1)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \text{--- (2)}$$

$$V_1 = i_b r_{\pi} = I_1 r_{\pi} \quad \text{--- (3)} \quad (2)$$

$$I_2 = \frac{V_2}{r_o} + \beta i_b = \beta I_1 + \left(\frac{1}{r_o}\right) V_2 \quad \text{--- (4)}$$

Comparing (1) & (3), $h_{11} = r_{\pi}$; $h_{12} = 0$ (2)

Comparing (2) & (4), $h_{21} = \beta$; $h_{22} = \frac{1}{r_o}$ (2)

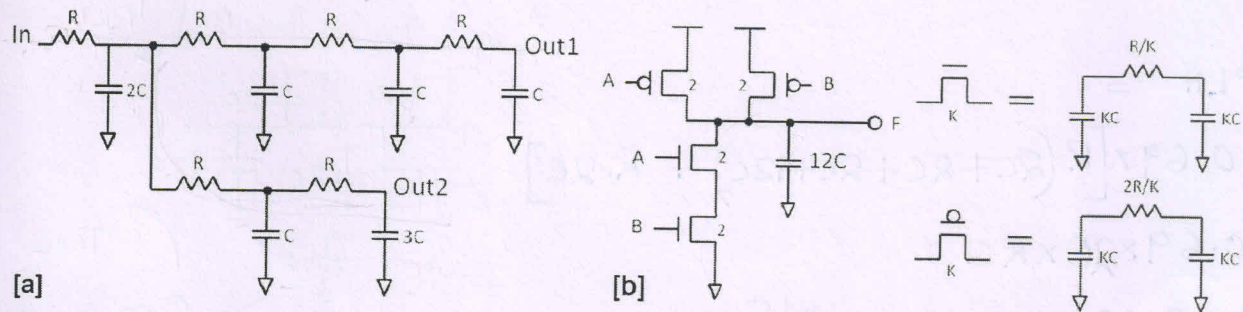


Fig. 1. (a) A RC network; (b) 2-input NAND gate and corresponding NMOS and PMOS transistor models.

Note: In delay calculation, use 'ln(2)' expression with $R = 100 \Omega$ and $C = 10 \text{ fF}$.

- (a) Given the RC network in Fig. 1(a), calculate the Elmore delay from **In** to **Out1** and from **In** to **Out2**. Which one is critical path? *delay in worst case for*
- (b) Calculate the T_{PLH} and T_{PHL} of a two input NAND gate shown in Fig. 1(b) using logical effort *assuming proportionality factor $\gamma = 1$ for submicron process. $\mu_n, \mu_p = 2:1$*
- (c) Calculate the T_{PLH} and T_{PHL} of the same two input NAND gate shown in Fig. 1(b) using Elmore delay.
- (d) Is the delay between part (b) and (c) different and why? Explain in one statement.

(a) In to Out1

$$\tau_1 = R(2C + C + C + C + C + 3C) + R(C + C + C) + R(C + C) + RC$$

$$= 15RC$$

$$td_1 = 0.69 \times 15 \times 100 \times 10 \times 10^{-15} = \underline{10.35 \text{ ps.}}$$

In to Out2

$$\tau_2 = R(2C + C + C + C + C + 3C) + R(C + 3C) + R(3C)$$

$$= 16RC$$

$$td_2 = 0.69 \times 16 \times 100 \times 10 \times 10^{-15} = \underline{11.04 \text{ ps.}}$$

* In to Out2 is the critical path.

(b) $T_{PLH} = T_{PHL} = t_{po}(p + f_g)$ When, $\gamma = 1$.

$t_{po} \Rightarrow$ Intrinsic delay of an inverter $= 3RC$.

$$P \Rightarrow \frac{C_{in}(\text{complex gate})}{C_{in}(\text{inverter})} = \frac{(2+2+2)C_{in}}{3C_{in}} = 2.$$

$$f \Rightarrow \frac{n+2}{3} = \frac{4}{3} \text{ for 2-input NAND.}$$

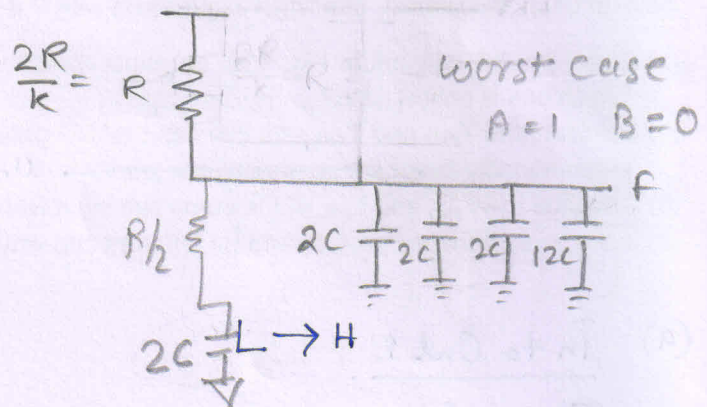
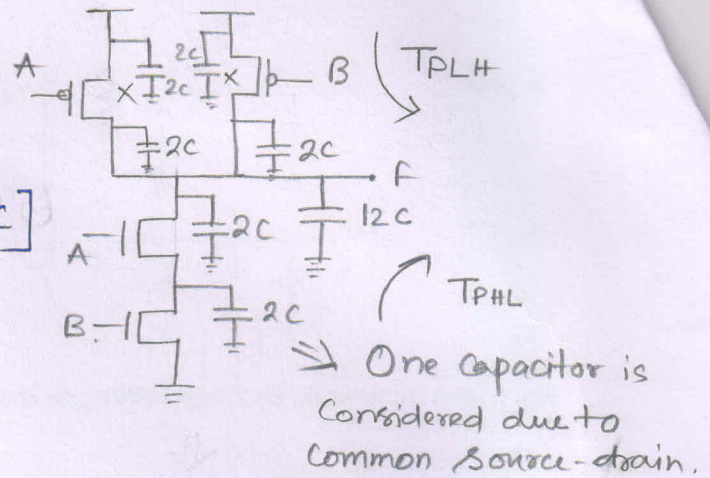
$$\text{Electrical effort } f = \frac{12C}{(2+2)C} = 3$$

$$T_{PLH} = T_{PHL} = 0.69 \times 3RC \times \left(2 + 3 \times \frac{4}{3}\right) = 12.42RC = 12.42 \times 100 \times 10 \times 10^{-15}$$

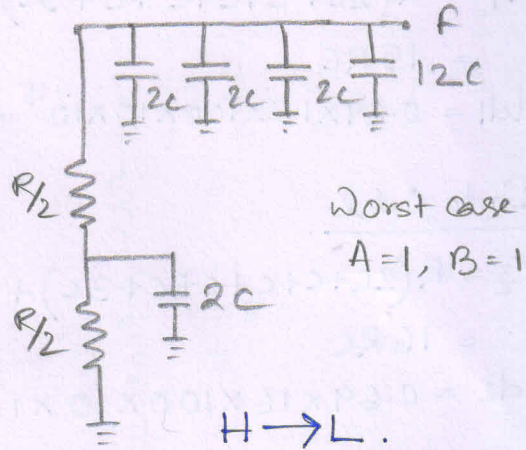
$$= \underline{12.42 \text{ ps.}}$$

(C) Delay from Elmore's model.

$$\begin{aligned}
 T_{PLH} &= \\
 &0.69 \times [R(2C + 2C + 2C + 12C) + R \cdot 2C] \\
 &= 0.69 \times 20 \times RC \\
 &= 0.69 \times 20 \times 100 \times 10 \times 10^{-15} \\
 &= \underline{13.8 \text{ ps.}}
 \end{aligned}$$



$$\begin{aligned}
 T_{PLH} &= \\
 &0.69 \times \left[\frac{R}{2} (2C + 2C + 2C + 2C + 12C) \right. \\
 &\quad \left. + \frac{R}{2} (2C + 2C + 2C + 12C) \right] \\
 &= 0.69 \times (10RC + 9RC) \\
 &= 0.69 \times 19 \times RC \\
 &= 0.69 \times 19 \times 100 \times 10 \times 10^{-15} \\
 &= \underline{13.11 \text{ ps.}}
 \end{aligned}$$



(d) ~~T_{PLH} is a match with T_{PLH} calculated from logical effort.~~
 T_{PLH} & T_{PHL} is higher in case of Elmore's model than Logical effort.

Reason: In T_{PHL} calculation, logical effort ignores the internal node capacitance.

- a) If CLOCK is launched at Node-A and Delay=25 ps. What is the Max. CLOCK frequency at which the system can be operated?

When clock is launched from Node-A and Delay is 25 ps!

Path-1 (Reg-A, Forward path and Reg-B) Positive skew experienced in this path

No race conditions ($15\text{ps} + 25\text{ps} < 65\text{ps}$)

can operate properly with a clock time period (min.) of

$$60\text{ps} + 85\text{ps} + 35\text{ps} - 25\text{ps} = 155\text{ps}$$

Path-2 (Reg-B, Backward path and Reg-A) Negative skew experienced in this path

No race conditions ($-5\text{ps} < 95\text{ps}$)

can operate properly with a clock time period (min.) of

$$70\text{ps} + 75\text{ps} + 30\text{ps} + 25\text{ps} = 200\text{ps}$$

So, the min. clock time period necessary is maximum of 155 ps & 200ps which is 200ps.

Clock frequency is 5 GHz

- b) What is the maximum tolerable value of *Delay*? For this you may have to consider two cases, CLOCK being launched at Node-A and Node-B separately.

When clock is launched at Node-A,

Path-1 will experience Positive Skew While Path-2 shall experience Negative skew.

Positive skew aggravates the problem of race conditions while negative skew eliminates them.

So, the limit on *Delay* is determined by path-1.

Max. tolerable value of *Delay* would be ($65\text{ps} - 15\text{ps}$) which is 50ps

When clock is launched at Node-B,

Path-2 will experience Positive Skew While Path-1 shall experience Negative skew.

Positive skew aggravates the problem of race conditions while negative skew eliminates them.

So, the limit on *Delay* is determined by path-2.

Max. tolerable value of *Delay* would be ($95\text{ps} - 20\text{ps}$) which is 75ps

Hence the max. tolerable value of *Delay* is 75 ps.

- c) What is advisable? Launching CLOCK from Node-A or Node-B? Justify. Clearly state the reasons.

ADVD

~~closed~~ open Bode

Q.1

$$V_{m1} - V_{m2} = V_{in}$$

$$I_{m1} = I_{m2} = I_{ss}/2 = 50 \mu A$$

$$C_L = 1 \text{ pF}, V_{ov} = 0.2 \text{ V}$$

(a)

OCMR $\downarrow$

$$\begin{aligned} \text{Upper limit} &\rightarrow (V_{DD} - |V_{GS1}|) = 2.5 - 0.6 = 1.9 \text{ V} \\ \text{lower limit} &\rightarrow V_{ov2} + V_{ov7} = 0.2 + 0.2 = 0.4 \text{ V} \end{aligned} \quad (2)$$

Output Signal Swing $\downarrow$

$$\begin{aligned} \text{Upper limit} &\rightarrow V_{DD} - |V_{TH}| = 2.5 - 0.4 = 2.1 \text{ V} \\ \text{lower limit} &\rightarrow V_{ov2} + V_{ov7} = 0.4 \text{ V} \end{aligned} \quad (2)$$

$$|\omega_{out}| = \frac{1}{R_{out} C_L} = \frac{1}{8 \text{ K} \times 5 \text{ pF}} = \frac{10^9}{40}$$

$$= 25 \times 10^6 \text{ rad/sec.}$$

$$= 26 \text{ Mrad/sec.} \quad (2)$$

$$g_{m1} = \frac{20 \times 50 \mu A}{0.2} = 500 \mu A/V$$

Small Sig. Current through M_1 & M_2 $\downarrow$

$$\begin{aligned} i_1 &= g_{m1} v_{in1} \\ &= 500 \mu A/V \times 5 \text{ mV} \\ &= 2.5 \mu A \end{aligned}$$

$$\begin{aligned} g_{m3} &= \frac{2 \times (I_{ss}/8)}{0.2} \\ &= 250 \mu A/V \\ &= 125 \mu A/V \end{aligned}$$

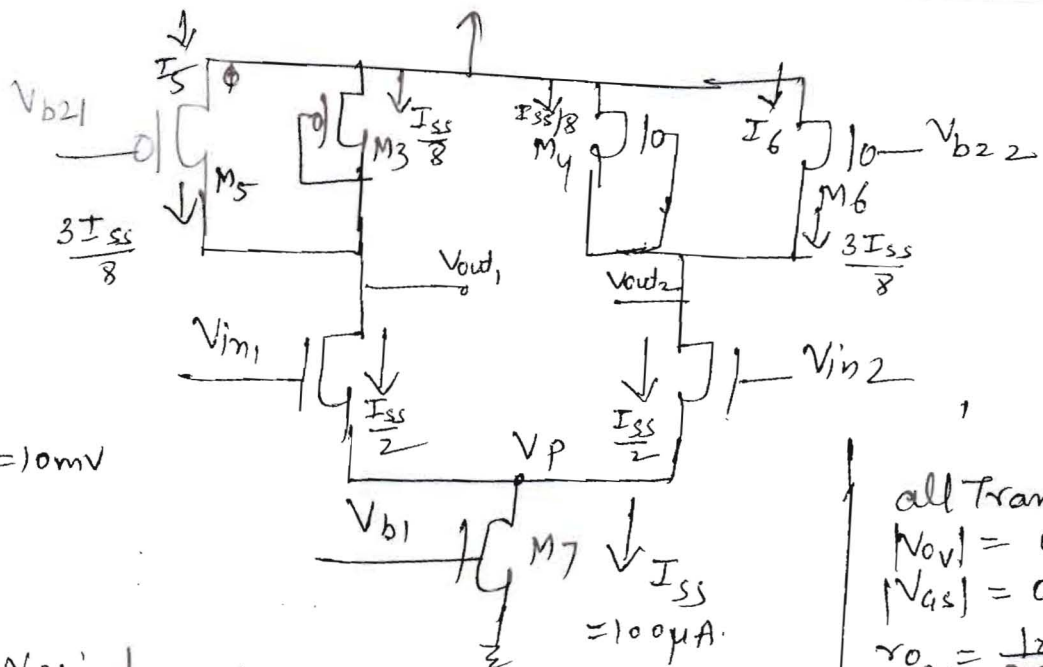
$$\frac{1}{g_{m3}} = 8 \text{ K}\Omega = 8 \text{ K}\Omega$$

$$\begin{aligned} r_{o1} &= \frac{1}{\lambda \left(\frac{I_{ss}}{2} \right)} \\ &= \frac{1}{0.05 \times 50 \mu A} \\ &= 400 \text{ K}\Omega \end{aligned}$$

$$\begin{aligned} R_{out} &= r_{o1} \parallel r_{o2} \parallel r_{o3} \parallel \frac{1}{g_{m4}} \\ &= 400 \text{ K} \parallel 53.3 \text{ K} \parallel 1.6 \text{ M} \parallel 8 \text{ K} \\ &= 8 \text{ K} \end{aligned}$$

$$r_{o6} = \frac{1 \times 8}{0.05 \times 300 \mu A}$$

$$\begin{aligned} r_{o4} &= \frac{8}{0.05 \times 100 \mu A} \\ &= 1.6 \text{ M} \end{aligned}$$



$$V_{in1} - V_{in2} = V_{in} = 10 \text{ mV}$$

V_{b21}, V_{b22} Varied such that

$$\begin{aligned} \frac{I_5}{5} &\Rightarrow \frac{4I_{ss}}{8} \rightarrow \frac{3I_{ss}}{8} \rightarrow \frac{2I_{ss}}{8} \\ \frac{I_6}{6} &\Rightarrow \frac{2I_{ss}}{8} \rightarrow \frac{3I_{ss}}{8} \rightarrow \frac{2I_{ss}}{8} \end{aligned}$$

S.NO	Parameter	Previous Value as in part (c)	New Value as in part (d)
1	I_5	$I_{ss}/8$	$\frac{4I_{ss}}{8} = I_{ss}/2$
2	I_6	$3I_{ss}/8$	$\frac{2I_{ss}}{8} = \frac{I_{ss}}{4}$
3	I_3	$I_{ss}/8$	$I_{ss}/8$
4	I_4	$I_{ss}/8$	$I_{ss}/8$
5	I_1	$I_{ss}/2$	$5I_{ss}/8$
6	I_2	$I_{ss}/2$	$3I_{ss}/8$
7	I_7	I_{ss}	I_{ss}
8	g_{m1}	$500 \mu\text{A/V}$	$625 \mu\text{A/V}$
9	g_{m2}	$500 \mu\text{A/V}$	$375 \mu\text{A/V}$
10	R_{out}	$8 \text{ k}\Omega$	$8 \text{ k}\Omega$

all Transistors

$$|V_{ov}| = 0.2 \text{ V}$$

$$|V_{gs}| = 0.6 \text{ V}$$

$$r_{o3,4} = \frac{1 \times 8}{0.05 \times I_{ss}} =$$

$$= 1.6 \text{ M}\Omega$$

$$r_{o5,6} = \frac{1 \times 8}{0.05 \times 300 \mu\text{A}} = 0.53 \text{ M}\Omega$$

$$g_{m3,4} = \frac{2 \times I_{ss}/8}{0.2} =$$

$$= \frac{2 \times 100 \mu\text{A}}{0.2 \times 8} =$$

$$= 125 \mu\text{A/V}$$

$$\frac{1}{g_{m3,4}} = 8 \text{ k}\Omega$$

$$r_{o1,2} = \frac{1 \times 2}{0.05 \times 100 \mu\text{A}} = 0.4 \text{ M}\Omega$$

$$g_{m1,2} = \frac{2 \times 50 \mu\text{A}}{0.2} = \frac{500 \mu\text{A}}{\text{V}}$$

$$R_{out} = r_{o1} \parallel \frac{1}{g_{m3}} \parallel \frac{1}{g_{m5}} \parallel \frac{1}{g_{m6}} \parallel \frac{1}{g_{m7}}$$

$$\approx 0.4 \text{ M}\Omega \parallel 0.53 \text{ M}\Omega \parallel 1.6 \text{ M}\Omega \parallel 8 \text{ k}\Omega$$

$$R_{out} \approx 8 \text{ k}\Omega$$

Modified Values $\rightarrow$

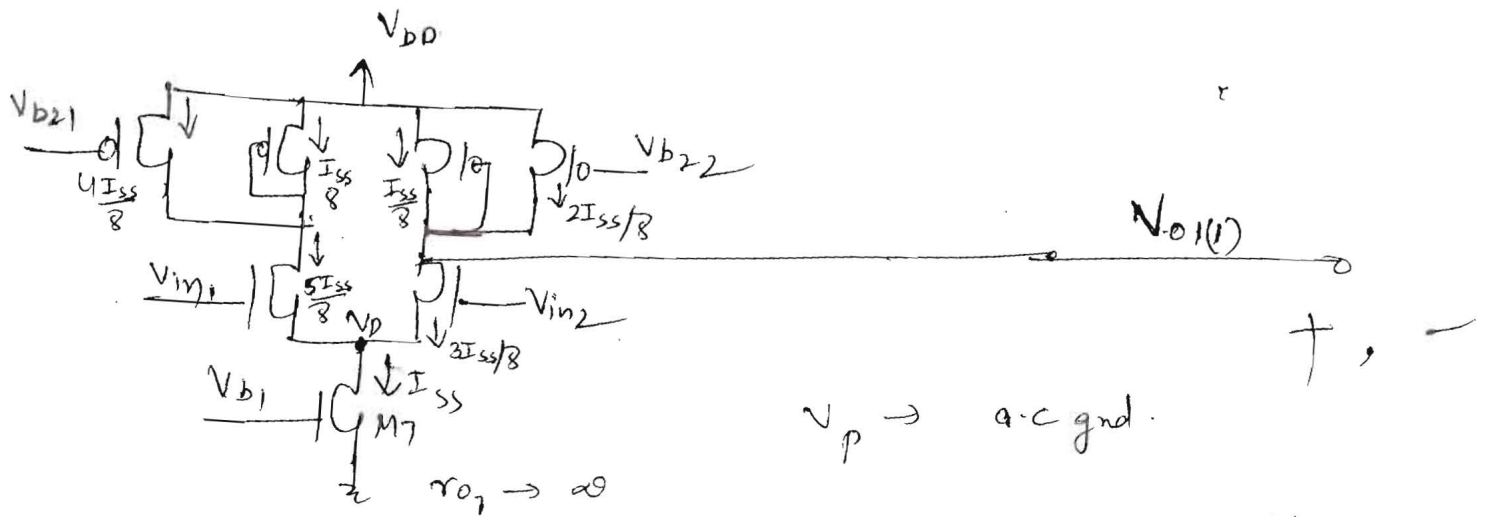
$$g_{m1} = \frac{2 \times 500 \mu\text{A}}{0.2} = 625 \mu\text{A/V}$$

$$g_{m2} = \frac{2 \times 300 \mu\text{A}}{0.2} = 375 \mu\text{A/V}$$

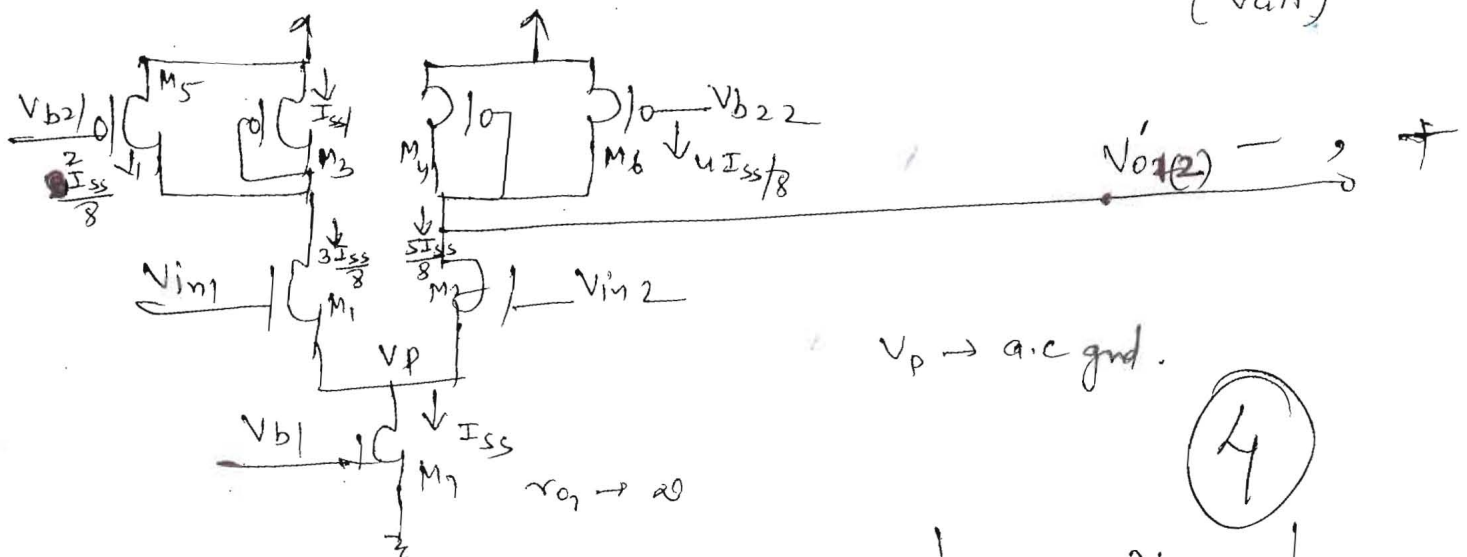
$$r_{o1} = 0.32 \text{ M}\Omega \quad r_{o2} = 0.53 \text{ M}\Omega$$

$$R_{out} = 8 \text{ k}\Omega$$

Variable Gain Amplifier $\rightarrow$



V_{out}
(VGA)



(4)

$$V_{out} \text{ of Variable Gain amplifier (VGA)} = |V_{o1(1)} - V_{o1(2)}|$$

when. $I_{1,2} = \frac{5I_{SS}}{8}$, $g_{m1} = \frac{625 \mu A}{V}$, $R_{out} = 8K$, $|V_{o1}| = 625 \mu \times 8K = 5$
 $I_{1,2} = \frac{3I_{SS}}{8}$, $g_{m1} = \frac{375 \mu A}{V}$, $R_{out} = 8K$, $|V_{o1}| = 3$

$$\left| \frac{V_{out}}{V_{in} \cdot V_{GA}} \right| = \left| \frac{V_{o1(1)} - V_{o1(2)}}{V_{in}} \right| = |5 - 3| = |2|$$

Hence. as I_S varies $4\frac{I_{SS}}{8} \rightarrow 3\frac{I_{SS}}{8} \rightarrow 2\frac{I_{SS}}{8}$
 I_6 varies $2\frac{I_{SS}}{8} \rightarrow 3\frac{I_{SS}}{8} \rightarrow 4\frac{I_{SS}}{8}$
 Simultaneously $\Rightarrow A_V$ (varies) (VGA) $+2$ to -2
 -2 to $+2$

[Note $\rightarrow$ Since. Both differential arms are having diff. bias current, so V_p will not be a.c. ground (proper). CMRR degraded.]

Ques. 2

(i) OP-AMP Transfer function:-

$$A = \frac{A_0 (1 - s/z)}{(1 + s/p_1)(1 + s/p_2)(1 + s/p_3)}$$

3

$$\angle A(j\omega) = -\tan^{-1}\left(\frac{\omega}{z}\right) - \tan^{-1}\left(\frac{\omega}{p_1}\right) - \tan^{-1}\left(\frac{\omega}{p_2}\right) - \tan^{-1}\left(\frac{\omega}{p_3}\right)$$

Given phase margin = 55° , $z = 10 \text{ GB}$, $p_3 = 4 \text{ GB}$

phase margin is given by $= 180^\circ + \angle A(j\omega)$

$$55^\circ = 180^\circ - \tan^{-1}\left(\frac{GB}{15GB}\right) - \tan^{-1}\left(\frac{GB}{p_1}\right) - \tan^{-1}\left(\frac{GB}{p_2}\right) - \tan^{-1}\left(\frac{GB}{4GB}\right)$$

$$= 180^\circ - \tan^{-1}\left(\frac{1}{15}\right) - \tan^{-1}\left(\frac{GB}{GB/A_0}\right) - \tan^{-1}\left(\frac{GB}{p_2}\right) - \tan^{-1}\left(\frac{1}{4}\right)$$

$$= 180^\circ - 3.8140 - 90^\circ - \tan^{-1}\left(\frac{GB}{p_2}\right) - 14.0362$$

$$55^\circ = 72.1498 - \tan^{-1}\left(\frac{GB}{p_2}\right)$$

$$\frac{GB}{p_2} = 0.3089 \quad \therefore p_2 = 3.24 \text{ GB}$$

$$p_2 = 32.405 \text{ MHz}$$

6

(ii) this case is a pole-zero cancellation case.

$$\text{p.m.} = 180^\circ - \left[\tan^{-1}\left(\frac{\omega}{p_1}\right) + \tan^{-1}\left(\frac{\omega}{p_3}\right) \right] \Big|_{\omega=GB}$$

$$= 180^\circ - 90^\circ - \tan^{-1}\left(\frac{1}{4}\right) = 75.96^\circ$$

5

Ques 3

for the bandgap reference circuit shown in figure

$$I_1 = I_2 = I_{D3} \quad (\text{note that } I_{D3} \neq I_3)$$

$$(2m) \quad I_2 = \frac{V_T \ln 10}{R_1} = I_{D3} \quad \left[V_T \ln 10 = 59.86 \text{ mV} \right]$$

$$I_{D3} = I_3 + I_{REF} + I_{Dm4} \quad (1)$$

$$\text{Output voltage} \quad V_{REF} = R_2 I_3 + V_{EB3} \quad (3m) \quad (11)$$

$$\text{from (1)} \quad I_3 = I_{D3} - I_{REF} - I_{Dm4}$$

$$I_{Dm4} = \frac{V_{REF}}{2R} \quad \& \quad I_{REF} = \frac{V_{REF}}{2R}$$

$$\text{Substituting the values in (11)} \quad V_{REF} = \left(\frac{V_T \ln 10}{R_1} - \frac{V_{REF}}{R} \right) R_2 + V_{EB3}$$

$$V_{REF} = \frac{R}{R+R_2} \left[\frac{R_2}{R_1} V_T \ln 10 + V_{EB3} \right] \quad (5m)$$

$$(ii) \quad \text{for ZTC} \quad \frac{\partial V_{REF}}{\partial T} = 0 \quad \left\{ \quad \frac{R_2}{R_1} = \frac{1.8}{0.087 \times \ln 10} = 8.98 \quad \right. \quad (5m)$$

$$R_2 = 8.98 R_1$$

$$(iii) \quad \text{if } R_1 = R = 1 \text{ k}\Omega, \quad R_2 = 10 \text{ k}\Omega \quad (2m)$$

$$V_{REF} = 0.118 \text{ V}$$

$$T_{COEFF} = \frac{1}{V_{REF}} \frac{\partial V_{REF}}{\partial T} = \frac{1}{0.118 \text{ V}} \left[0.203 \text{ mV} \times \frac{1}{11} \right] = 156.39 \text{ ppm/}^\circ\text{C} \quad (3m)$$

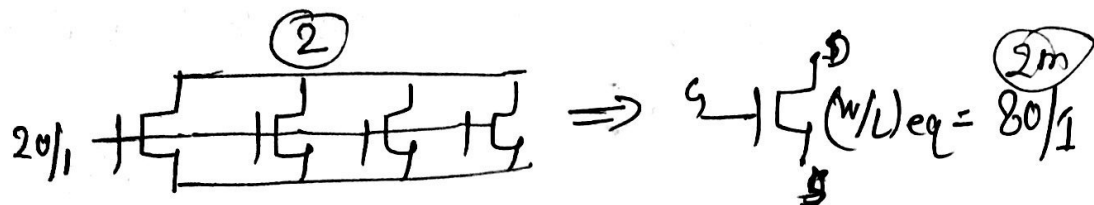
$$T_{COEFF} = 156.39 \text{ ppm/}^\circ\text{C}$$

ADVD MIDSEM -2019

Solution

Part B Solutions

Q. 1(a)



$$C_{SB} :- (10 + 20 + 10) \times C_{JSW} + (2 \times 10) C_{JSW} + (10 + 20 + 10) C_{JO} + 3 \times 10 \times 20 C_{JO}$$

$$= 100 C_{JSW} + 600 C_{JO}$$

$$= 10 + 300 = 310 \text{ fF} \quad (3m)$$

$$C_{dB} = (10 + 10) C_{JSW} + (10 + 10) C_{JSW} + 2 \times 10 \times 20 \times C_{JO}$$

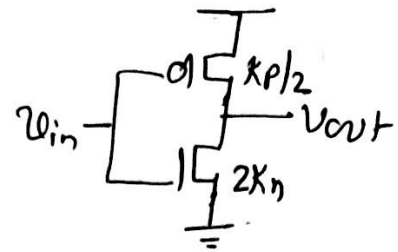
$$= 40 C_{JSW} + 400 C_{JO}$$

$$= 204 \text{ fF} \quad (3m)$$

Ques 1(b) Given $V_{TH} = 1.25V$, $V_{DD} = 2.5V$

The equivalent inverter:-

$$V_{TH(NOR-2)} = \frac{V_{TH} + \frac{1}{2} \sqrt{\frac{K_P}{K_N}} (V_{DD} - V_{TH})}{1 + \frac{1}{2} \sqrt{\frac{K_P}{K_N}}}$$

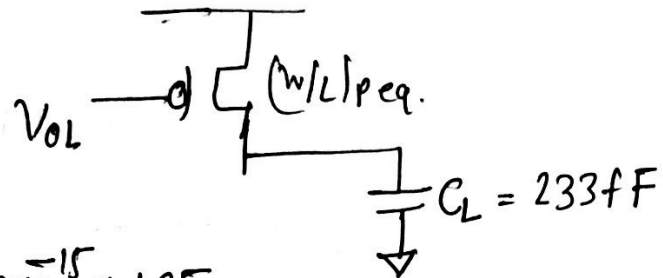


$$0.425 \sqrt{\frac{K_P}{K_N}} = 0.85 \quad \sqrt{\frac{K_P}{K_N}} = 2 \quad (2m)$$

$$(W/L)_P = 12(W/L)_N \Rightarrow (2m)$$

$$T_{PLH} = 200ps$$

for average current method



$$T_{PLH} = 200ps = \frac{233 \times 10^{-15} \times 1.25}{I_{avg}} \quad (1)$$

$$\begin{aligned} I_{avg} &= \frac{1}{2} \left[i_c(V_{in}=V_{OL}, V_{out}=V_{OL}) + i_c(V_{in}=V_{OL}, V_{out}=V_{50\%}) \right] \\ &= \frac{1}{2} \left[\frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right)_{Peq} (2.5 - 0.4)^2 + \mu_p C_{ox} \left(\frac{W}{L} \right)_{Peq} \left[2.5 - 0.4 - \frac{1.25}{2} \right] 1.25 \right] \\ &= \frac{1}{2} [88.2 (W/L) + 73.75 (W/L)] = 80.975 (W/L) \mu A \quad (2m) \end{aligned}$$

Pulling the value in (1) $(W/L)_{Peq} = 17.983 \approx 18$

$$\therefore (W/L)_P (PMOS) = 361 \quad (2) \quad (W/L)_N = 3/1 \quad (2)$$

for $L = L_{min} = 0.25$ $(W/L)_P (PMOS) = 4.5/0.25$, $(W/L)_{NMOS} = 0.75/0.25$

$$\alpha \rightarrow 1 = \gamma_4 \times 3/4 = 3/16$$

$$\begin{aligned} P &= (233 \times 2.5^2 \times 10 \times 10^{-9}) \times 3/16 = (25.825 \mu W) \times \frac{3}{16} = 4.5625 \times \frac{3}{16} \mu W \\ &= 2.73 \mu W \quad (4) \end{aligned}$$

Que 2(a)

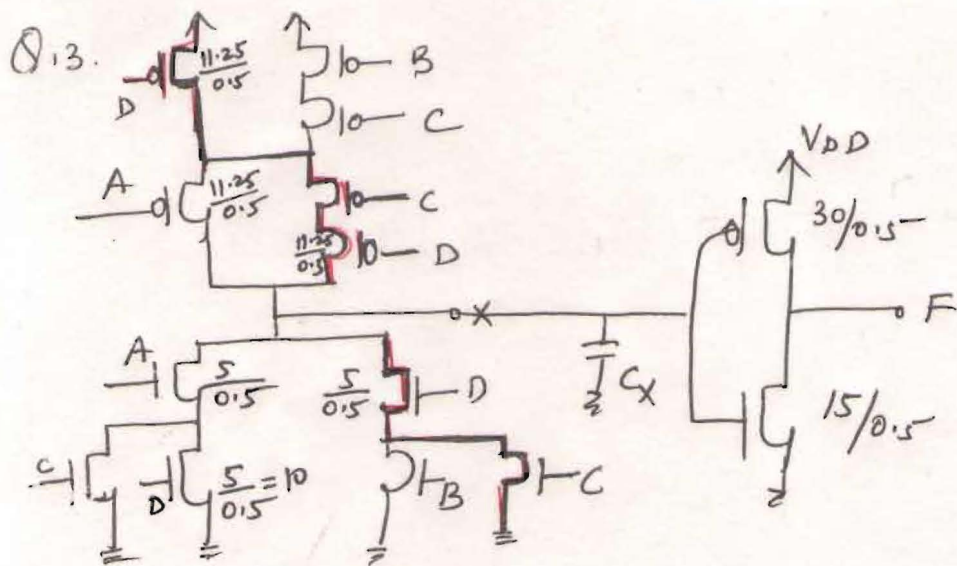
C-terms.	Expression	Calculation.	Value H → L	Value L → H
C _{PD1}	$L_{DN} \times W_n \times C_{ox}$ $+ \frac{1}{2} C_{ox} W_n L$ $L \rightarrow H$ $L_{DN} \times W_n \times C_{ox}$	$H \rightarrow L$ $0.052 \times 0.375 \times 6$ $+ \frac{1}{2} \times 6 \times 0.375 \times 0.25$ $L \rightarrow H$ $0.052 \times 0.375 \times 6$	0.398 fF	0.117 fF
C _{PD2}	$L_{DP} \times W_p \times C_{ox}$ $L \rightarrow H$ $L_{DP} \times W_p \times C_{ox}$ $+ \frac{1}{2} C_{ox} W_p L$	$H \rightarrow L$ $0.045 \times 1.125 \times 6$ $L \rightarrow H$ $0.045 \times 1.125 \times 6 + \frac{1}{2} \times 6 \times 1.125 \times 0.25$	0.304 fF	1.14785 fF
C _{DB1}	$k_{effn} \times C_j \times (W_n \times Y)$ $+ k_{esn} \times C_{jsn} \times 2(W_n \times Y)$ $+$	$H \rightarrow L$ $0.57 \times 2 \times 0.0375 \times 0.8$ $+$ $0.61 \times 0.28 \times 2(0.375 + 0.8)$ $L \rightarrow H$ $0.79 \times 2 \times 0.375 \times 0.8$ $+$ $0.81 \times 0.28 \times 2(0.375 + 0.8)$	0.74 fF	1.00 fF
C _{DB2}	$k_{effp} \times C_j \times (W_p \times Y)$ $+ k_{esp} \times C_{jsp} \times 2(W_p \times Y)$	$H \rightarrow L$ $0.79 \times 1.9 \times 1.125 \times 0.8$ $+$ $0.86 \times 0.22 \times 2(1.125 + 0.8)$ $L \rightarrow H$ $0.5 \times 1.9 \times 1.125 \times 0.8$ $+$ $0.7 \times 0.22 \times 2(1.125 + 0.8)$	2.08 fF	1.45 fF

Que. 2(b) Process engineer will face punch-through for all 'L' which are $\leq 2\mu m$

$$W_d = \sqrt{\frac{2 \times 11.8 \times 8.85 \times 10^{-14}}{1.6 \times 10^{19}}} \left(\frac{1}{10^{15}} \right) \text{ cm} \quad (2m)$$

$$= 9.6 \times 10^{-5} \text{ cm or } 0.96 \mu m$$

$$\therefore L = 2 \times 0.96 = 1.92 \mu m \quad (2m)$$



W.C	Path	$(\frac{W}{L})_{peq}$
PU	A-C-D	$\frac{1}{3}(\frac{W}{L})_p$
	B-C-D	$\frac{3}{8}(\frac{W}{L})_p$
	A-B-C	$\frac{1}{3}(\frac{W}{L})_p$

P.D	Path	$(\frac{W}{L})_{Neq}$
	A-D	$\frac{1}{2}(\frac{W}{L})_N$
	D-B	
	D-C	
	A-C	

all W_p same

all W_N same

highlighted Worst Case paths. (W.C)

$$(\frac{W}{L})_{peq} = [D-C-D]_{path} = \frac{1}{3}(\frac{W}{L})_p$$

$$(\frac{W}{L})_{Neq} = [C-D] = \frac{1}{2}(\frac{W}{L})_N = \frac{1}{2} \times 10 = 5 \quad (1)$$

(a) Boolean Expression $X = \overline{A(C+D) + D(B+C)} \quad (3)$

(b) Equivalent Inverter - Worst Case PU/PDN Network

$$(\frac{W}{L})_{peq,wc} = \frac{1}{3}(\frac{W}{L})_p \quad (1)$$

$$\frac{\mu_n C_{ox}}{\mu_p C_{ox}} = \frac{120}{40} = 3$$

$$(\frac{W}{L})_{Neq,wc} = \frac{1}{2}(\frac{W}{L})_N \quad (1)$$

For Symmetric Pull up / Pull down Network $\frac{W_p}{W_N} = \frac{\mu_n}{\mu_p} \Rightarrow$ unit inverter

In Symm. Worst Case Equivalent Inverter $\rightarrow$

$$\frac{(\frac{W}{L})_{peq,wc}}{(\frac{W}{L})_{Neq,wc}} = 3 \quad (1)$$

$$\frac{1}{3}(\frac{W}{L})_p = 3 \times \frac{1}{2}(\frac{W}{L})_N \Rightarrow (\frac{W}{L})_p = \frac{9}{2}(\frac{W}{L})_N = \frac{9}{2} \times 5 = \frac{45}{2}$$

$$\text{if } L_p = 0.5 \mu m \Rightarrow W_p = \frac{45}{2} \times 0.5 \mu m = 11.25 \mu m \quad (1)$$

$$t_{PHL} = 0.7 R_{Neq.wc} C_X \quad (1)$$

C_X is load
Capacitance at Node X
ignoring Parasitics

$$C_X = C_{gpmos} + C_{gnmos} \quad (\text{of Inverter II stage})$$

$$= W_P L C_{ox} + W_N L C_{ox} = (30+15) \mu m \times 0.5 \mu m \times C_{ox} \quad ; \quad C_{ox} = 6 fF/\mu m^2$$

$$= 135 fF \quad (2)$$

$$R_{Neq.wc} = \frac{13 K \Omega}{(W/L)_{Neq.wc}} = \frac{13 K}{5} = 2.6 K \quad (2)$$

$$t_{PHL} = 0.7 \times 2.6 K \times 135 fF = 245.7 psec = 0.245 ns \quad (2)$$

(c) $g_A = g_B = g_C = g_C$

$$g_A = \frac{16.25}{20} = 0.8125 \quad (1)$$

$$h = \frac{C_X}{C_{A,B,C...}} = \frac{30+15}{16.25} = 2.769 \quad (1)$$

$$P = \frac{C_{pg}}{C_{pinv}} P_{inv} = \frac{2(11.25+5)}{15+5} = 1.625 \quad (1)$$

$$d = gh + p \quad (1)$$

$$= \left(\frac{16.25}{20} \right) \left(\frac{45.9}{16.25} \right) + \frac{2 \times 16.25}{20}$$

$$= \frac{9}{4} + \frac{16.25}{10}$$

$$d = 2.25 + 1.625 = 3.875 \text{ units} \quad (1)$$

$$d = 0.775 F04 \text{ delay}.$$

$$1 F04 \text{ delay} = 5 \text{ units}.$$

time constant τ of technology node $\downarrow$

$$d = \frac{d_{abs}}{\tau} \quad (1)$$

$$\tau = \frac{0.245 ns}{3.875} = \frac{245.7}{3.875} ps.$$

$$\boxed{\tau = 63.4 ps} \quad (1)$$